

Week 4 Interim Report

Project: KAIM 5 – EthioMart E-commerce Data Extractor

Branch: week4-data-ingestion

Date: 2025-06-22

1. Channels Selected

- ZemenExpress
 - nevacomputer
 - meneshayeofficial
 - ethiobrandcollection
 - Leyueqa
 - sinayelj
 - Shewabrand
 - helloomarketethiopia
 - modernshoppingcenter
 - qnashcom
- (...and 12 others as per assignment)

2. Ingestion Workflow

1. Auth & Setup

- Loaded TG_API_ID & TG_API_HASH from .env.
- Initialized Telethon client session (kaim_week4_session).

2. Channel Crawling

- Iterated each channel via `client.iter_messages(limit=10000)`.
- Extracted id, date, sender, text fields.
- Gracefully skipped invalid usernames.

3. Output

- Saved combined JSON to preprocessing/
raw_messages_<timestamp>.json.

3. Preprocessing Steps

1. Normalization

- Stripped Amharic diacritics (unicode U+135B-U+1363).
- Converted Ethiopian numerals (፩ – ፩፱) to Latin digits (1-9).

2. Cleaning

- Removed URLs, collapsed line breaks, normalized whitespace.

3. Tokenization

- Simple `str.split(' ')` into tokens.

4. Output

- Saved token lists to preprocessing/
processed_<timestamp>.json.

4. CoNLL Labeling

- **Entities:** PRODUCT, PRICE, LOCATION (BIO tags).
- Labeled **30** sample messages in `labeling/labeled_data.conll.txt`.
- Ran `labeling/validate_conll.py` to ensure correct format.